1. Clinical Question & Core Hypothesis

Clinical Question

Among MIBC tumors, why does apparent immunogenicity from MMR deficiency (MSH6/MSI) fail to translate into checkpoint■inhibitor benefit in some cases—i.e., what microenvironmental features blunt neoantigen-driven immune responses? 2) Tier 1 paths used: I built on the TGF-β fibrogenic/immune■suppressive axis (core path 1) and the MSH6/MSI immunogenicity axis (core path 2), because together they pose a testable mechanism where stromal fibrosis may antagonize MMR-driven neoantigen benefit. I considered SARIFA (core path 3) but deprioritized it as a primary anchor to keep the mechanistic claim focused. 3) Tier 2 support: AI■based biomarker and immunotherapy■biomarker paths justify using imaging (TSR) as a scalable readout of stromal programs and support translational assays; metabolic/systemic markers (HDL■C, IGF■1 etc.) provide orthogonal signals that could modulate or confound associations and are therefore listed as secondary modifiers. 4) Choice of endpoint: I selected therapy_response (immunotherapy response proxy) because the central puzzle is mechanistic (why neoantigen load does not always produce clinical benefit) and because MSI/MSH6 biology is most actionable in the context of checkpoint therapy. 5) Deliberate exclusions: I excluded PSA and stool methylation concepts from the core hypothesis because they lack direct mechanistic linkage to MIBC stromal–immune interplay; SARIFA and systemic metabolic markers were moved to secondary_modifiers to avoid overfitting the primary claim and keep the hypothesis experimentally tractable?

Working Hypothesis

In MSH6-deficient bladder cancers, a high stromal-to-tumor ratio driven by TGF-β signaling creates a fibrotic, immunosuppressive microenvironment that blunts neoantigen–specific T cell responses despite a high mutational/neoantigen burden. Clinically, this TSR-high, TGF-β–active, MSH6-deficient phenotype will exhibit reduced response to immune checkpoint blockade compared with TSR-low tumors, and this poor response may be improved by combining anti–TGF-β–targeted therapy with immunotherapy.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TGF-β, MSH6
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TGF-β: TGF-β fibrogenic/immune■suppressive axis (core path 1) and the MSH6/MSI immunogenicity axis (core path 2), because together they pose a testable mechanism where stromal fibrosis may antagonize MMR-driven neoantigen benefit.

MSH6: MSH6/MSI immunogenicity axis (core path 2), because together they pose a testable mechanism where stromal fibrosis may antagonize MMR-driven neoantigen benefit.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	SARIFA	Supporting modifier from knowledge graph
Secondary	MSI (phenotype)	Supporting modifier from knowledge graph
Secondary	AI-based biomarker integration	Supporting modifier from knowledge graph
Secondary	HDL-C/IGF-1 (systemic metabolic signals)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Retrospective clinical analyses in MIBC cohorts with documented MSH6/MSI status and ICI outcomes to correlate TSR (imaging/histology) with response rates;

Prediction 2: Preclinical studies in MSH6-deficient models assessing CTL infiltration and tumor control under PD-1 blockade with and without anti–TGF-β therapy;

Prediction 3: mechanistic assays (pSMAD2/3 signaling, CAF markers, ECM deposition) and transcriptomic profiling to link TSR with TGF-β activity and immune signatures;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- SARIFA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MSI (phenotype): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.